

ROS 2 TurtleBot3 Lab Report

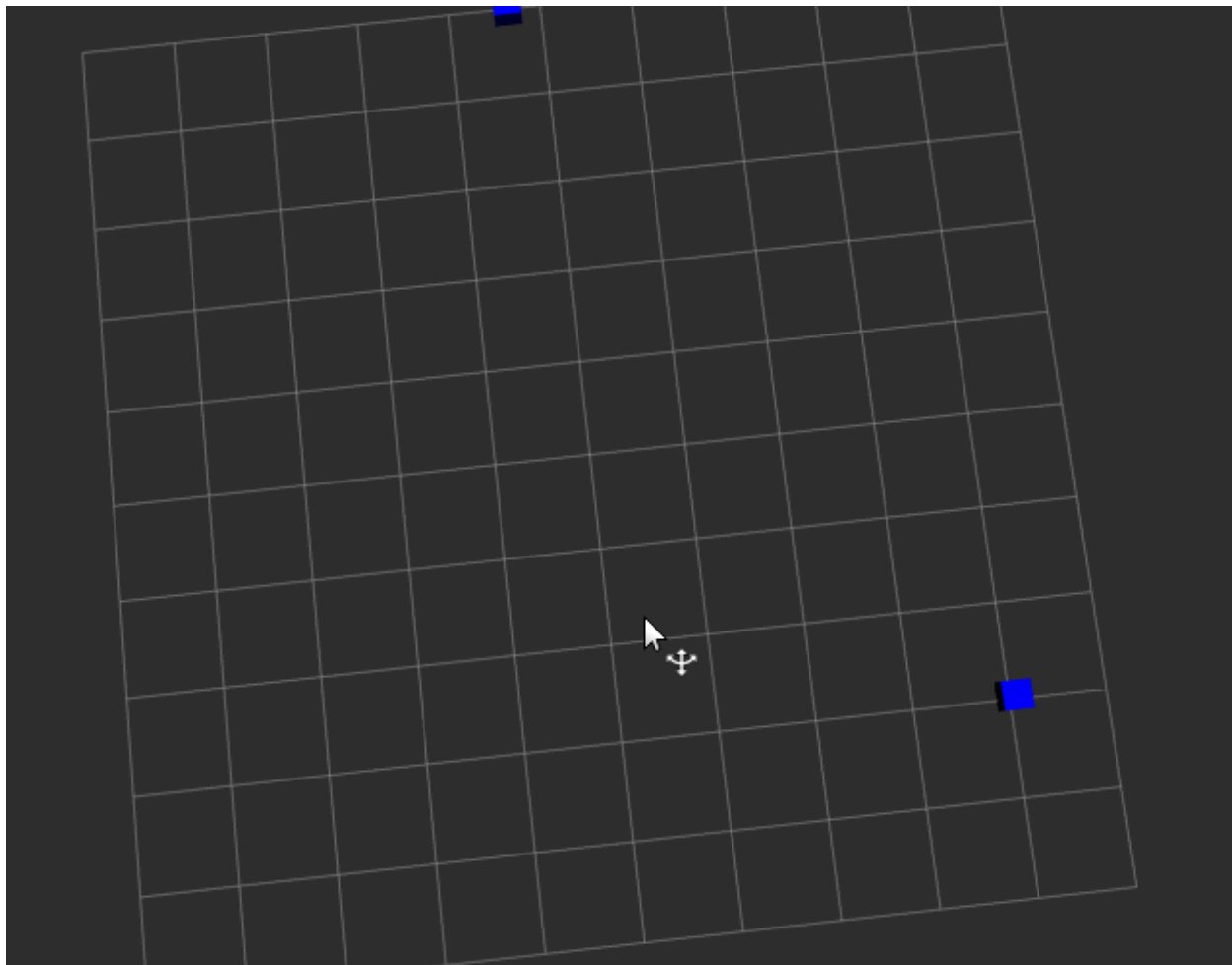
Muhammad Ahmad 25280065

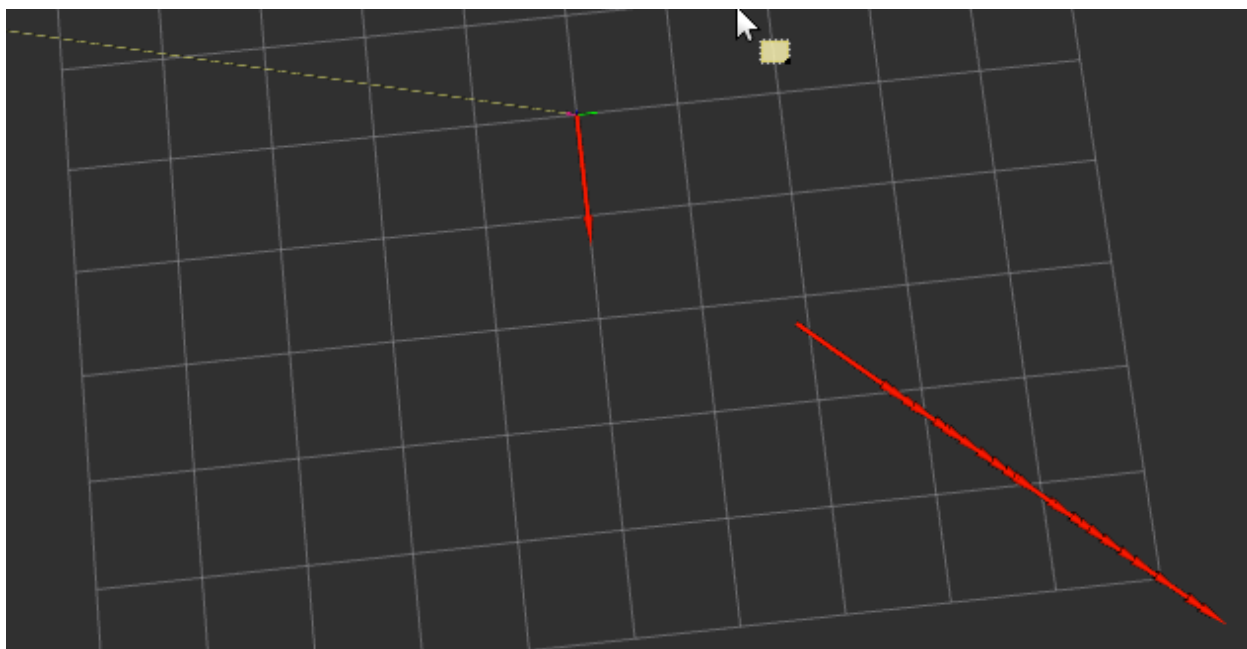
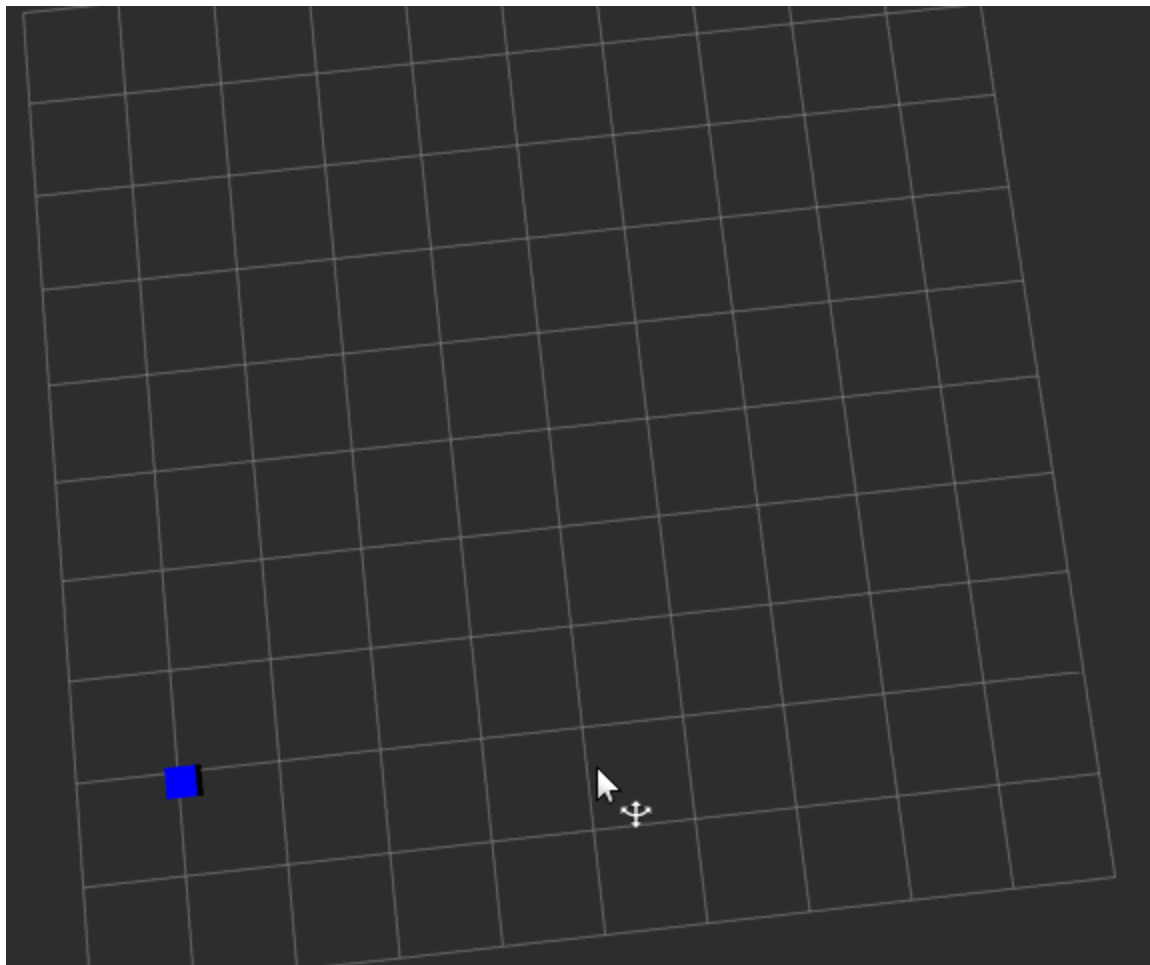
Task 1: Setup and Basic Motion

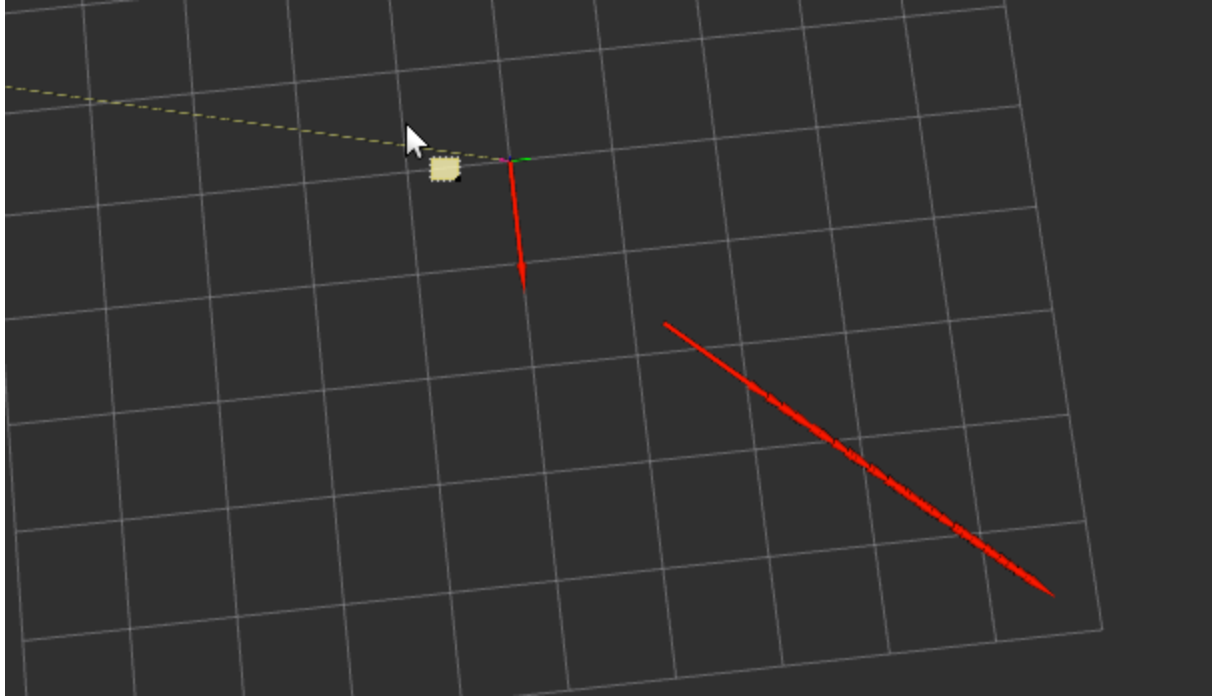
In this task, I set up the TurtleBot3 simulation and verified that the robot could move using teleoperation. I opened RViz to visualize the robot, TF frames, and LaserScan data.

Then I created a Python script to move the robot in a circular path with a radius of 5 meters by publishing velocity commands. The robot successfully moved in a circle in the simulation.

After that, I implemented the go-to-goal script. The robot used odometry data to know its position and orientation, converted quaternion to yaw, and moved from the start position to the goal point (5,5).



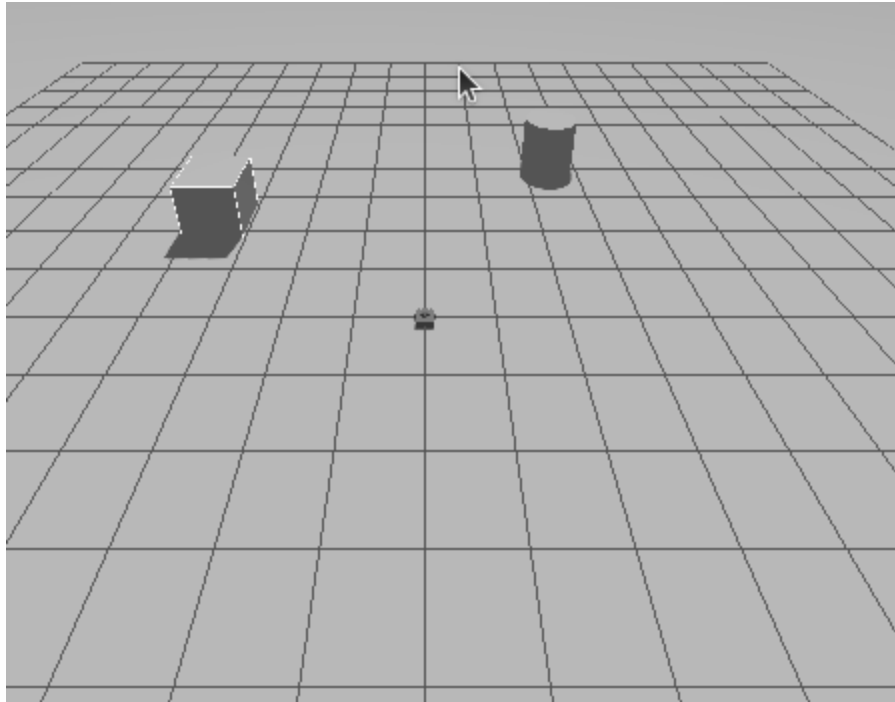


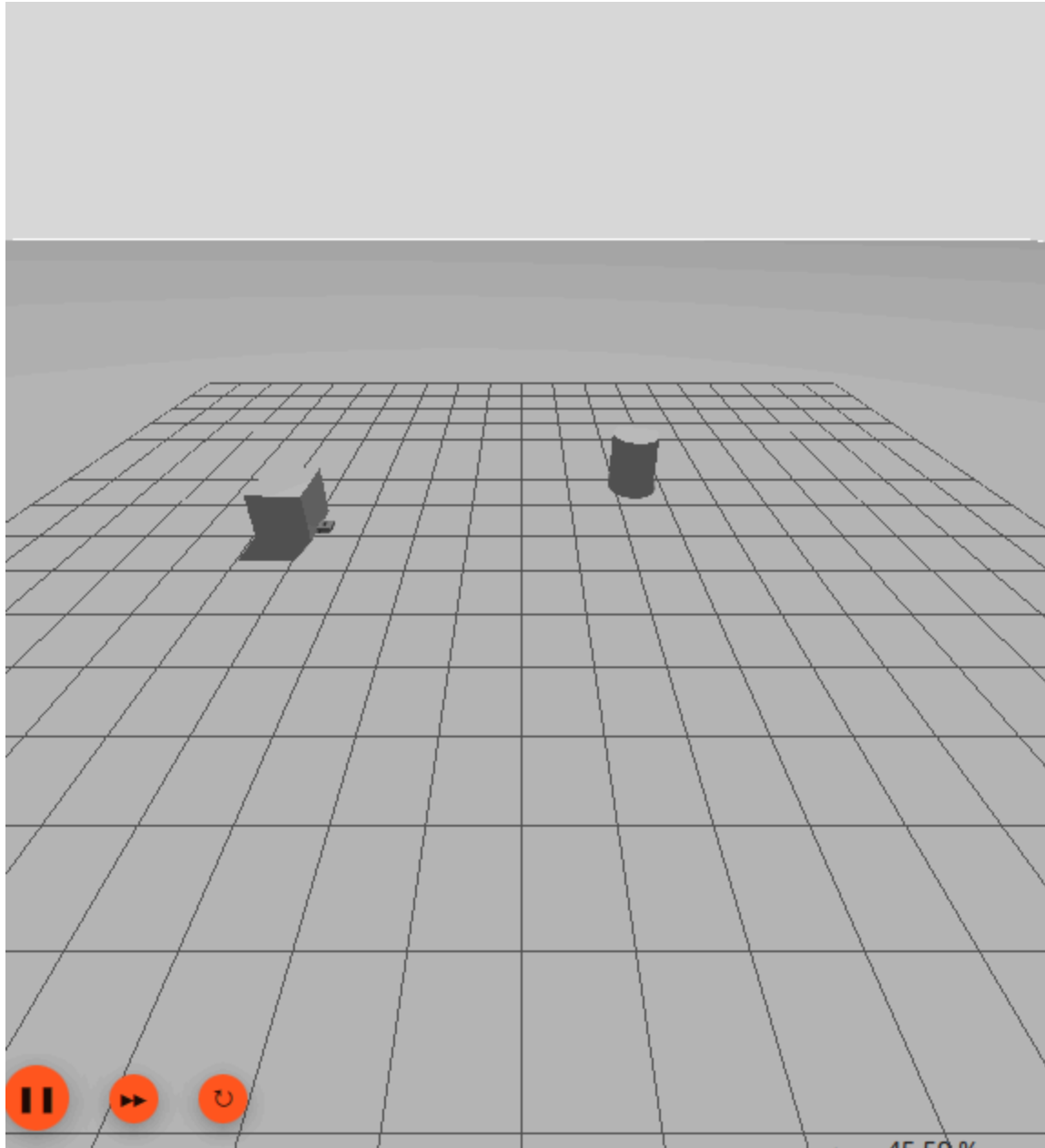


Task 2: LaserScan Data

In this task, I wrote a script to subscribe to Lidar data and print the number of valid readings, the closest point, and the farthest point with their angles.

When the robot moved closer to the object, the closest distance value decreased, which showed that the sensor was working correctly.





```
===== LIDAR INFO =====
[INFO] [1771168825.968384151] [turtlebot3_laserscan]: Valid readings: 150
[INFO] [1771168825.968552573] [turtlebot3_laserscan]: Closest point: 0.12 m at angle 354.81 deg
[INFO] [1771168825.968730121] [turtlebot3_laserscan]: Farthest point: 0.53 m at angle 286.65 deg
[INFO] [1771168825.968890335] [turtlebot3_laserscan]: =====
[INFO] [1771168826.170399014] [turtlebot3_laserscan]:
===== LIDAR INFO =====
[INFO] [1771168826.170564561] [turtlebot3_laserscan]: Valid readings: 150
[INFO] [1771168826.170723983] [turtlebot3_laserscan]: Closest point: 0.12 m at angle 1.00 deg
[INFO] [1771168826.170867530] [turtlebot3_laserscan]: Farthest point: 0.56 m at angle 286.65 deg
[INFO] [1771168826.171010868] [turtlebot3_laserscan]: =====
[INFO] [1771168826.393265447] [turtlebot3_laserscan]:
===== LIDAR INFO =====
```